

Design and Control of Integrated Chromatography Column Sequences

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To increase the productivity in biopharmaceutical production, a natural step is to introduce integrated continuous biomanufacturing which leads to fewer buffer and storage tanks, smaller sizes of integrated unit operations, and full automation of the operation. The main contribution of this work is to illustrate a methodology for design and control of a downstream process based on integrated column sequences. For small scale production, for example, pre-clinical studies, integrated column sequences can be implemented on a single chromatography system. This makes for a very efficient drug development platform. The proposed methodology is composed of four steps and is governed by a set of tools, that is presented, that makes the transition from batch separations to a complete integrated separation sequence as easy as possible. This methodology, its associated tools and the physical implementation is presented and illustrated on a case study where the target protein is separated from impurities through an integrated four column sequence. This article shows that the design and control of an integrated column sequence was successfully implemented for a tertiary protein separation problem. © 2017 American Institute of Chemical Engineers Biotechnol. Prog., 33:923–930, 2017

Keywords: continuous bioprocessing, integrated column sequences, straight through processing, autonomous downstream processing, multiple column control

Introduction

To increase the productivity in biopharmaceutical production a natural step is to introduce integrated continuous biomanufacturing which leads to fewer buffer and storage tanks, smaller sizes of integrated unit operations, and full automation of the operation. The challenges in continuous upstream and downstream processing are different and all has to be addressed in an integrated context.^{1,2} Continuous upstream processes forces the downstream processes to handle continuous feed streams.^{3,4} Integration of several processing steps has previously been discussed.^{5–7} The dominating separation techniques in biopharmaceutical production are based on

fixed bed chromatography which by nature results in periodic operations causing the amount and concentration of the output to vary periodically.⁸ This can in principle be handled in two ways, a series of continuous separation steps with buffer tank between the steps to handle and reduce the variation^{9,10} or by a chain of integrated synchronized separation steps without buffer tanks. This second concept is called integrated column sequence (ICS).¹¹

In ICSs, where the buffer tanks between the columns are removed, the separation train of chromatography steps is very tightly connected and the pool from one column is directly loaded on the next column. In a tightly connected process, like an ICS, the operation of the complete process has to be well synchronized. The capacity of each step has to be selected to give a well-balanced system and to avoid bottlenecks there is a need for complete automation of the downstream processing.¹²

Additional Supporting Information may be found in the online version of this article.

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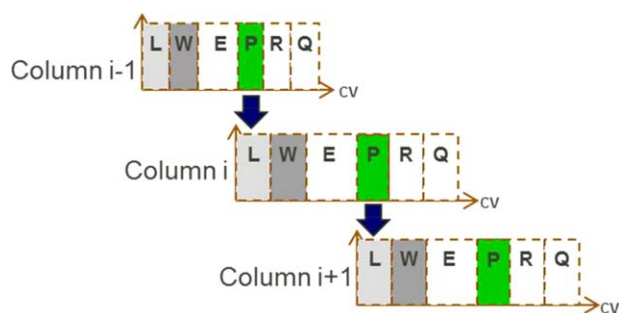


Figure 1. Integrated column sequence with tight and synchronized connection between the steps.

In biopharmaceutical industry new production technology has to be introduced very early in product and process development due to regulation issues and time pressure in the development project. Integrated continuous biomanufacturing needs to be studied already in discovery and pre-clinical tests where the first small scale production of product candidates are done. In early development, small amounts of a large set of drug candidates needs to be produced. The separations are often performed in a set of steps making the small scale production a labor intensive work. The introduction of continuous processing forces the technology to be scalable from lab and pre-clinical pilot production to production for late clinical tests and commercial scale. A scalable technology means a methodology for equipment sizing, configuration, and operation at different scale of production.

Integrated chromatography column sequences, ICS, will be addressed in this work with a methodology for design and control at laboratory scale. The methodology is based on the idea from Tiainen et al.¹³ and is implemented on commonly available equipment to be easily accepted. The concept is illustrated on a case study in lab scale. An external controller for multi column automation is presented together with a methodology for controller design. The result is an autonomous small scale mini factory that can make the drug development more cost efficient. This is illustrated in a case study where a protein is separated from other proteins using an ICS.

Theory

The full integration of a sequence of chromatography columns forces the eluting pool to be directly loaded on the next column, see Figure 1. In general, this is not possible without mobile phase adjustments. Therefore, in ICS systems every second step is a step for adjusting the mobile phase, for example, desalting, pH adjustment or buffer exchange. In this work, gel filtration columns are used between every bind-and-elute column to move the eluting pool from elution condition in the previous column to the loading condition in the next column. A final gel filtration column is added in the end of the sequence to change the last product pool to suitable storage conditions.

The design of the sequence creates a set of constraints on the connected columns, namely; (i) the pool volume from column i have to be equal or smaller than the maximal load volume for column $i + 1$, (ii) the amount of protein in the pool from i have to be equal or less than the capacity of column $i + 1$, (iii) the volumetric flow rates have to be the same in pool i and load $i + 1$. These three constraints have to be fulfilled for every connection and there are infinite

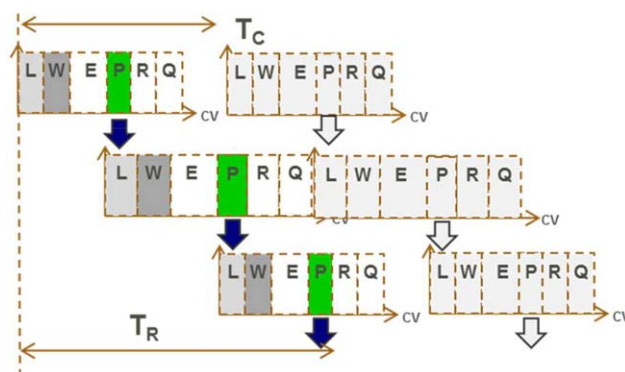


Figure 2. Integrated column sequence on a multiple setup configuration.

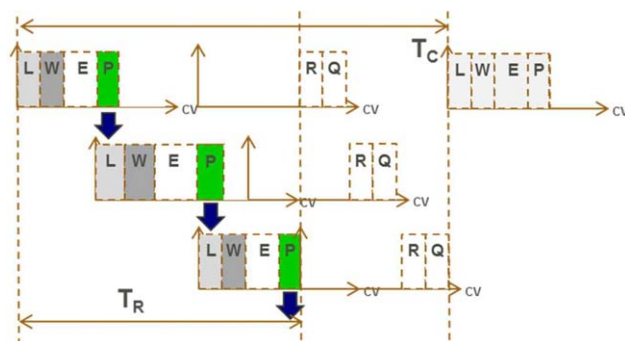


Figure 3. Integrated column sequence on a single setup configuration, nICS.

number of solutions but the most interesting ones are the ones that minimize column volumes or minimize production time for the complete sequence.

The ICS can be implemented on different number of setups, from a single ICS setup (nICS), illustrated in this work, to a multiple ICS setup (mICS) where every step in the sequence is run on an individual machine, see Figure 2. A special case is the twin setup configuration, where two identical machines are connected. The first machine runs odd steps and the second machine runs even steps, giving the twin-setup a cycle time that is close to the product residence time, but the twin-setup will not be discussed here.

The product residence time, T_R , will be the same for both the nICS and mICS but the possibility to regenerate the column and start a new cycle naturally gives the multiple setup a higher production rate. The cycle time will be limited by the longest individual step, T_C , for a multiple setup configuration.

The single setup configuration means that every step is parked after elution and regenerated after the final product have left the sequence, T_R , see Figure 3. This gives the single setup long cycle time, T_C , almost the same as all steps together. Still this is attractive in lab scale due to the limitations of available equipment.¹³

Assuming that the separation in each chromatography step is well defined and each phase, its liquid volume and flow rate, are specified in column volumes, it is possible to scale the separation by assuming that the separation is unchanged if the residence time is constant.^{14,15} The amount eluted from the previous step has to be loaded on to the next step. For an adsorption column, the capacity has to be larger than the amount eluted from the previous column, constraint ii

Table 1. Separation Specifications for the Capture and Polishing Steps in Normalized Units (Volume: Column Volumes, Flow Rate: Column Volumes Per Minute)

	Load	Prepool	Pool	Postpool
Capture	9 CV 2 CV/min	0.5 CV 2 CV/min	4 CV 2 CV/min	5 CV 2 CV/min
Polish	0.83 CV 0.33 CV/min	1.83 CV 0.33 CV/min	0.5 CV 0.33 CV/min	5 CV 0.33 CV/min

Prepool is all necessary steps between the load and the pool such as wash and start of elution. Postpool is all steps after the pool such as regeneration, cleaning in place, and equilibration of the column.

above, and for a gel filtration step, the load volume of the step has to be larger than the eluted volume, constraint i. This creates inequality constraints between the columns and their sizes. In some situations, the eluting flow rate has to be decreased because of flow rate limitations on the loading phase of small high capacity adsorption columns, constraint iii. For each individual column two variables have to be found, the column volume the elution flow rate. The number of variables for the optimization is $2N$, where N is the number of steps, that is, N column volumes and N elution flow rates. The number of constraints is two for each column connection, that is, $3N - 2$ for a complete nICS system. The elution flow rate at the end is given and finally a reference for calculation is needed. Requirement on processed feed, amount produced product or a given column volume are examples of different calculation references. Finally an objective for the optimization has to be defined. The most common ones are the product residence time or the total size of all columns. The optimization problem above results in a linear programming problem.^{16,17}

In the case of continuous upstream processes, like perfusion reactors, the feed rate and the protein load are almost constant. This forces the ICS to start with a continuous capture. One way to implement this is to introduce two parallel capture columns where the first is continuously loaded and the other one connected to the ICS and eluted to the next column.¹¹ This process design forces the capacity of the capture columns to be so large that the loading takes longer time than the complete cycle of the ICS. This is the case for continuous upstream processes with low expression levels. The capture column constraint is easily introduced into the optimization problem. The constant feed is used as the basis for calculation and an extra constraint is introduced, where the product residence time has to be less than two capture loading times, $T_R < 2 T_C$.

Based on the solution of the optimization the synchronized procedure to run the ICS system is known. Before the automatic generation of the control procedure all important time delays and mixing, due to piping, pumps, and mixers have to be specified. The final result is a temporal and state dependent procedure that can be translated to a finite state machine description with events and actions, which is used to control the chromatography system.

Materials and Methods

Case study

In the case study, a target protein is separated from other proteins in two chromatography steps, one capture step and one polish step. The ICS separation is performed using single setup integrated column sequence, nICS system, which requires two extra gel filtration columns. The resulting nICS system configuration is composed of four columns; (1) the

capture step, (2) desalting column step, (3) gradient ion exchange step, and (4) formulation column step. The capture step is used to concentrate the target in a small pool. In this case study, the capture is an ion exchange column with a step elution. Step two is a desalting step where the pool is desalted to adsorb on the third step. The polish step, step three, is based on linear gradient elution on a cation exchanger where the target is pooled between two major impurities. The last step in an ICS system is a formulation step where the buffer in the final product pool is exchanged for storage or formulation.

The specification on step one and three is found in Table 1 and batch chromatogram data have been used as a starting point in the design of the complete nICS. The gel filtration columns used in step two and four have common specification. The specification of the gel filtration columns are a maximal load volume of 0.3 CV, a maximal pool volume of 0.3 CV, and the maximal flow rate of 2 CV/min.

Materials

All experiments were conducted using an ÄKTA Pure system from GE Healthcare (Buckinghamshire, United Kingdom) with a column valve, an extra loop valve and two extra versatile valves. The methodology presented is not limited to this chromatography system. Any chromatography system with support for a communication protocol such as OPC could be used. The following columns from GE Healthcare were used in the case study; prepacked HiTrap SP HP 1 mL column, prepacked Resource S 6 mL column and five prepacked G-25 HiTrap Desalting columns. PEEK tubing with an inner diameter of 0.75 mm was used to connect pumps, valves, sample loop, column, and sensors.

Ribonuclease A from bovine pancreas, cytochrome c from horse heart, lysozyme from chicken egg white, sodium monobasic phosphate, sodium dibasic phosphate, and sodium chloride were purchased from Sigma Aldrich (St. Louis, MO). Acetone was purchased from Merck (Darmstadt, Germany). The low salt buffer contained 20 mM sodium phosphate at pH 7. The high salt buffer contained 20 mM sodium phosphate and 500 mM sodium chloride at pH 7.

Experimental Setup. The setup is configured as in Figure 4 with an injection valve V1, a super loop SC, a column valve V3, a UV detector, a conductivity detector, an outlet valve V6, an extra loop valve V4, two extra versatile valves V2 and V5. This setup can handle eight columns without reconfiguration based on the column valve and loop valve, where each have four column positions and one by-pass position. The two versatile valves, one before and one after the column sections of V3 and V4, are used to select which valve is first and forces the other valve to be second. This means that the sensors are always measuring after the first column section in the sequence and the outlet valve is only

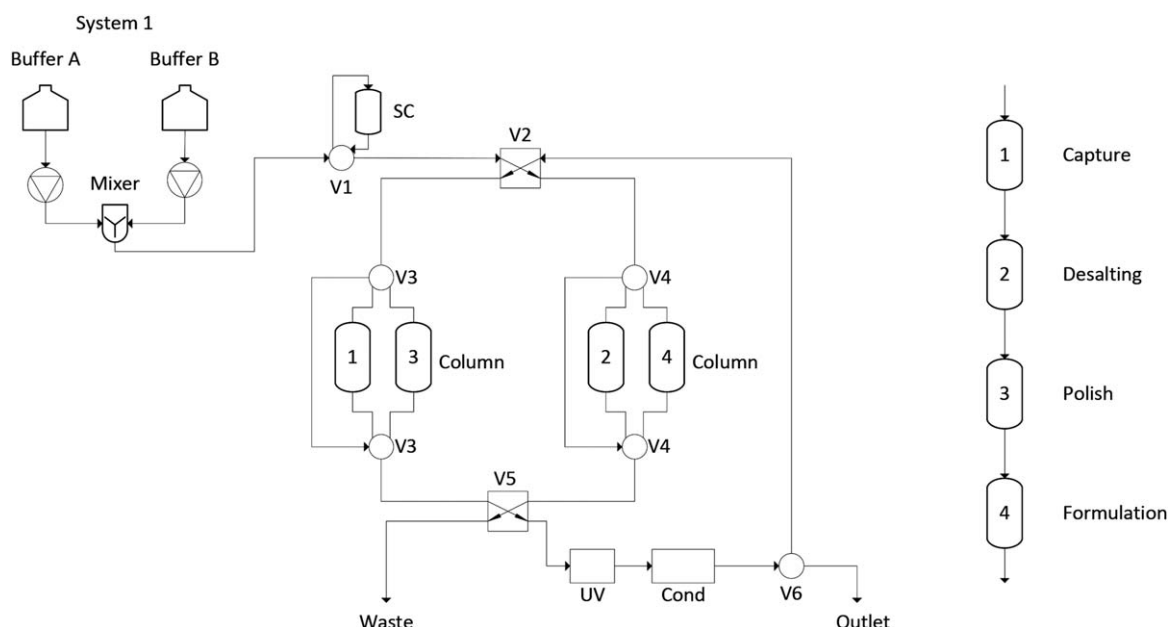


Figure 4. Single setup integrated column sequence configuration, nICS.

used for final product pooling. The outlet from the second column section is eluted to waste from V5.

Design method

The design of the ICS is performed using an optimization of the case study configuration based on the theory presented above. The column sizes and flow rates are found by minimizing the objective, in this case minimal column sizes. Even though the optimization problem is linear,¹⁷ we selected a general constraint optimization method due to implementation aspects. The optimization problem is solved in the Python programming language¹⁸ using the Sequential Least Squares Programming method¹⁹ in the SciPy package.²⁰ This method is based on sequential least square programming for a scalar objective and multiple decision variables and allows both inequality and equality constraints.

External controller

The ICS is configured on the chromatography system with the accompanying control system (Unicorn 6.4).²¹ The predefined way to use the control system is to run a single column in an open loop setup. The control system makes it possible to interact with the chromatography system through an OPC interface^{22,23} for communication with other software, see Figure 5. This feature is exploited in the development of an external controller for multi column separation, implemented in Python. Through the interface Python can read and write to the control system on a real time basis making it possible to read sensor measurement and change valve positions and pump flow rates. The external controller is not limited to one setup but can communicate with multiple setups and other additional equipment.

A research software called Orbit²⁴ has been developed to run the chromatography using a scripting approach. In Orbit, a main class for the system is loaded which contains methods to connect to and disconnect from the system, see Figure 6. An instance of the system class contains instances to all defined units of the system, such as pumps, valves, columns and sensors.²⁵ The user interacts with the chromatography

system by wrapping the methods of the OPC server interface that can change positions of valves and get measurements.

There are two ways to run Orbit. The basic way is to let the experimental setup be controlled by sending out various commands in a script written by the user specifying which commands that should be executed at defined time points. Orbit can also utilize information about piping, sensors, and unit operations to calculate the appropriate timings and control sequence. In this work, the scripting method was used for the batch runs and the automatic approach was used for the nICS run. The code that was used to run the nICS chain can be found in the Supporting Information.

Results and Discussion

The main result in this work is the presentation and verification of methodology that can take a description of a downstream process and automatically design both the corresponding ICS and its control system. The methodology consists of four steps; (1) nICS design, (2) controller design, (3) nICS verification, and (4) nICS operation.

nICS design

The design is based on the original separation specification, see Table 1, for the capture and the purification steps together with general design specifications for the additional gel filtrations steps. The design of the sequence starts with chromatograms from the individual steps. In Figure 7, a run of the capture step is shown where the column is loaded and eluted. After a short wash, the proteins are eluted with a salt step and a suitable pool is selected. The next step is the polishing step shown in Figure 8, where a salt gradient is applied to an ion-exchange column to achieve purification.

Based on the selected size of the capture step, in this case 1 mL, all the other column sizes and flow rates can be found through optimization. The selected objective in this case is the minimization of the total column volumes. The optimization result is seen in Table 2. The column volumes are found together with the pool flow rate to fulfill the connection

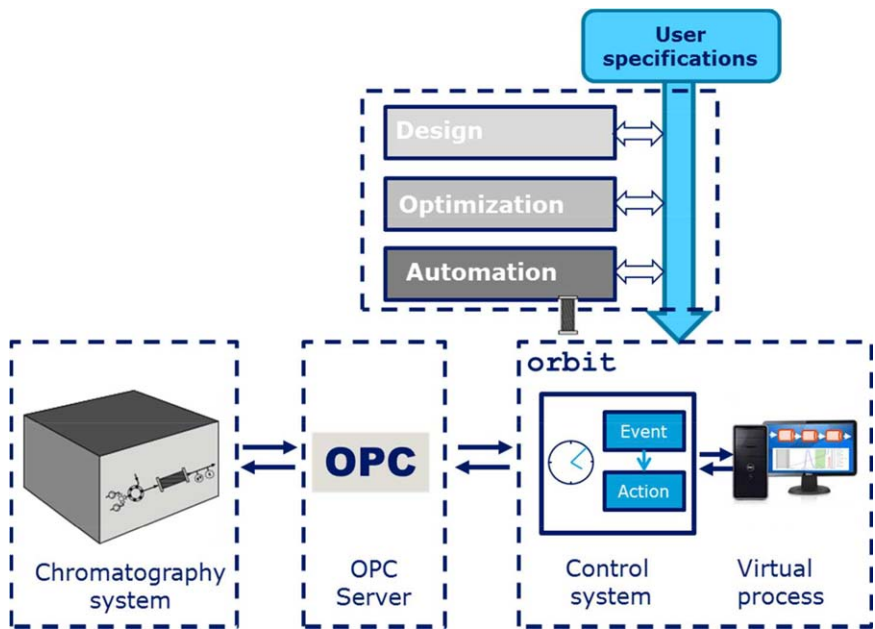


Figure 5. Orbit controller is a program developed in Python enabling execution of chromatography through scripting. Orbit instructions are created from user specifications through design, optimization and automation. Orbit executes its commands through a connection to an OPC to control the chromatography system.

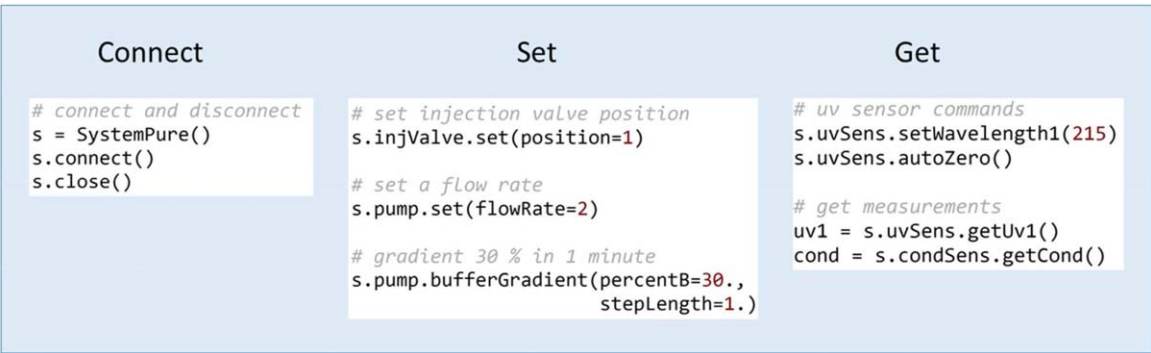


Figure 6. A few examples of commands available in the controller Orbit.

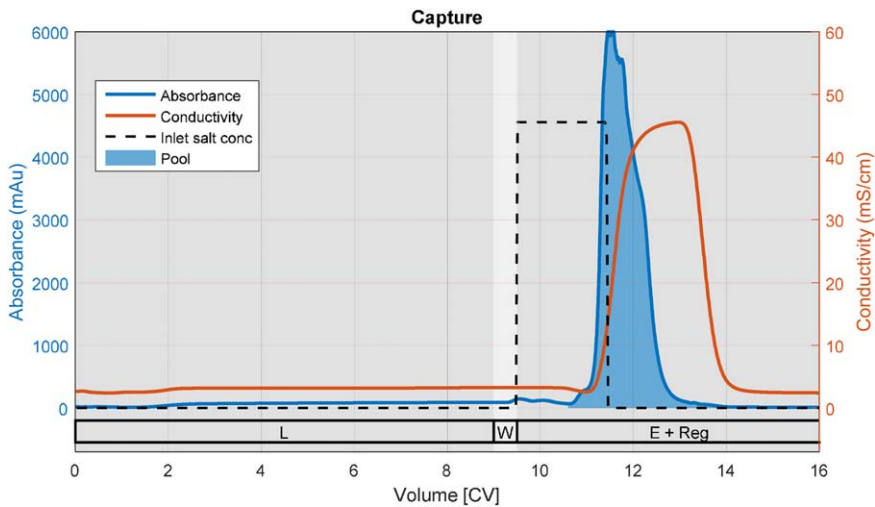


Figure 7. The chromatogram of the capture step. The column is loaded until saturated, then washed and eluted with a salt pulse and the pool (blue area) contains a mixture of all proteins. The bars at the bottom indicate the phase of the batch run, where load (L), wash (W), elution (E), and regeneration (Reg) is denoted.

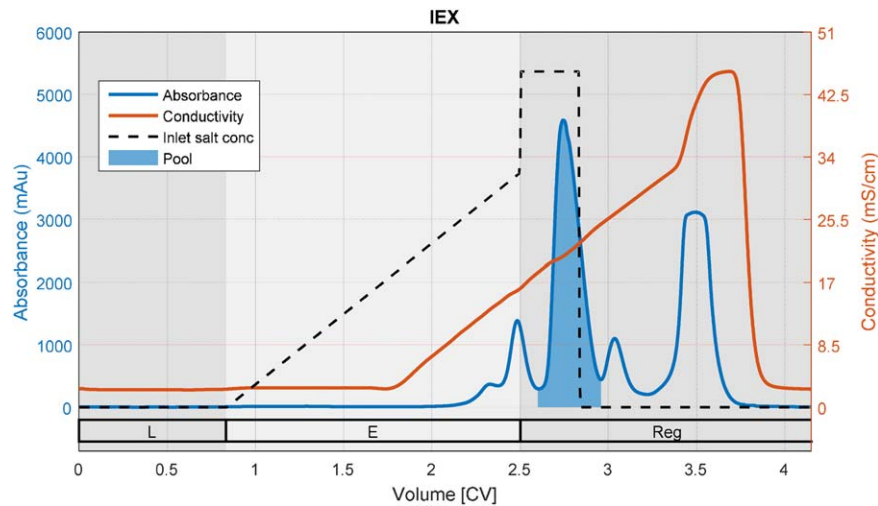


Figure 8. The chromatogram of the polishing step.
An ion exchange chromatography is used for the separation and the product is pooled (blue area). The bars at the bottom indicate the phase of the batch run, where load (L), elution (E), and regeneration (Reg) is denoted.

Table 2. The Optimal nICS Design Results

	Size	Load	Prepool	Pool	Postpool
Capture	1 mL	9 CV 2 CV/min	0.5 CV 2 CV/min	4 CV 2 CV/min	5 CV 2 CV/min
Desalting	14.1 mL	0.28 CV 0.14 CV/min	–	0.3 CV 0.12 CV/min	1 CV 2 CV/min
Polish	5.98 mL	0.7 CV 0.33 CV/min	1.83 CV 0.33 CV/min	0.5 CV 0.32 CV/min	5 CV 0.33 CV/min
Formulation	10.5 mL	0.29 CV 0.18 CV/min	–	0.3 CV 2 CV/min	1 CV 2 CV/min

requirement discussed above. In this case study, the pool flow from the large desalting step is reduced to fit the load flow rate for the polishing column.

After the optimization, the actual physical columns have to be packed and prepared. Due to packing variations, the sizes of the packed columns does not match the optimal sizes, it can therefore become necessary for a modifying optimization of the flow rates based on the set of actual packed column sizes. The optimization problem is complemented with equality constraints on the column volumes resulting in flow rates fulfilling all constraints.

Controller design

Important time delays in piping are calculated based on the selected flow rates in all different flow paths during a complete nICS procedure. The size and length of all pipes in the configuration are known and based on the valve positions and flow rate the time between two columns can be calculated. These time delays are added to the nICS design from step 1 generating a complete nICS scheme. In Orbit, the specific configuration of the chromatography system is defined including all pipes and their volumes. The complete design is the base for the automatic generation of the finite state machine description for Orbit, the Orbit specification.

nICS verification

The Orbit specification can be studied virtually through simulations to check if the nICS design have been done

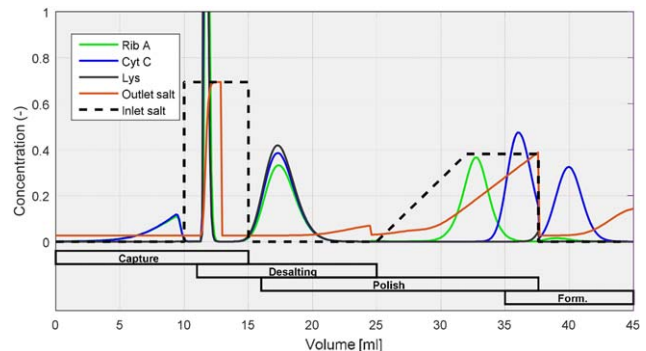


Figure 9. Simulation of four column case study based on an Orbit specification.

The simulation illustrate the capture column 0–15 mL, desalting step 11–25 mL, polishing column 16–37 mL, formulation step 35–45 mL, and with the final product pool in 3.5 mL. The regeneration and equilibration are not seen in the simulation.

correctly and generated a proper procedure. A simulation based on the generated controller sequence is illustrated in Figure 9. The response from all four steps in the ICS is seen and the final product pool with the target protein is collected in 3.5 mL. The simulation tool is also implemented in Python using the Assimulo²⁶ package that wraps CCode numerical method²⁷ to solve differential-algebraic equation systems. The simulation used a kinetic dispersive model¹⁵ with a Langmuir mobile phase modulator isotherm²⁸ further described by Ojala et al.²⁹ It was calibrated using the batch experiments shown in Figures 7 and 8. After simulation,

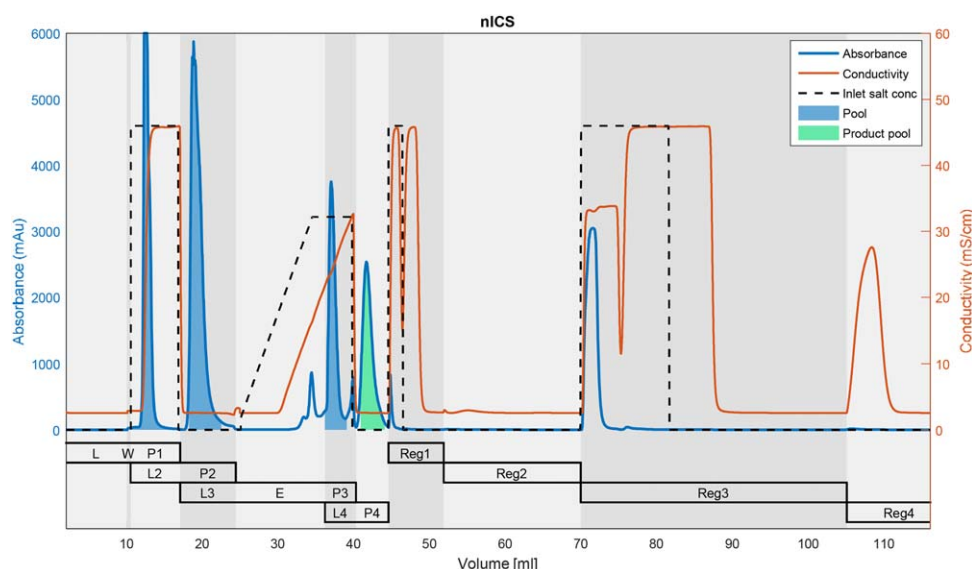


Figure 10. A complete separation of the case study nICS system using the multi column controller Orbit.

The pools from the three first steps are filled in blue and the product pool is filled in green. The bars at the bottom indicate the phase of the nICS run, where load (L), wash (W), pool (P), elution (E), and regeneration (Reg) is denoted.

Orbit is run on the actual system. The first complete separation is performed step by step to make it possible for the user to check and correct the complete nICS procedure. After the verification, the operation can start by performing a set of batches or just run the nICS system for a given time period.

nICS operation

After the nICS design has been optimized and verified, a run is performed on the chromatography system and the resulting chromatogram is shown in Figure 10. The capture step is seen as the first part of the chromatogram. The pool is eluted at around 13 mL and is directly fed to the gel filtration column to separate the salt. In the gel filtration column, the peak elutes at around 19 mL and is pooled to the polishing step. The salt that is separated in the gel filtration column starts to elute just before the switch at the end of the P2 section. The proteins are then bound to the stationary phase of the polishing step and separated with a gradient. The pool from the polishing step is eluted at around 38 mL and is directly fed to formulation step where salt once again are removed. The product pool is taken at around 42 mL (green area). Now the regeneration of the columns remains. For the capture step and polishing column, high salt concentration is used to remove proteins retained within the column, followed by an equilibration of the columns so that it is ready for the next cycle. In the regeneration of the capture column, there is a dip in the conductivity which is caused by remaining buffer with low salt concentration in the pipes from the polishing step. In the regeneration of the polishing step, the peak that elutes is the remaining hardly bound component lysozyme.

Conclusions

The main contribution in this work is to illustrate a methodology for design and control of a downstream process based on ICSs, called ICS in the article. For small scale production, like in pre-clinical studies, ICS implemented on a

single chromatography system, is a very interesting alternative for efficient drug development. The proposed methodology is composed of four steps; (1) ICS design, (2) the controller design, (3) ICS verification, and finally (4) ICS operation. This chain of steps is governed by a set of tools that makes the transition from a set of separation steps to a complete ICS as easy as possible. The methodology, the associated tools and the physical implementation, is illustrated on a case study where the target protein is separated from impurities through an integrated four column sequence.

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Literature Cited

1. Konstantinov KB, Cooney CL. White paper on continuous bioprocessing may 20-21 2014 continuous manufacturing symposium. *J Pharm Sci.* 2014;104:813–820.
2. Zydny AL. Perspectives on integrated continuous bioprocessing—opportunities and challenges. *Curr Opin Chem Eng.* 2015; 10:8–13.
3. Godawat R, Konstantinov K, Rohani M, Warikoo V. End-to-end integrated fully continuous production of recombinant monoclonal antibodies. *J Biotechnol.* 2015;213:13–19.
4. Warikoo V, Godawat R, Brower K, Jain S, Cummings D, Simons E, Johnson T, Walther J, Yu M, Wright B, McLarty J, Karey KP, Hwang C, Zhou W, Riske F, Konstantinov K. Integrated continuous production of recombinant therapeutic proteins. *Biotechnol Bioeng.* 2012;109:3018–3029.
5. Ojala F, Sellberg A, Hansen TB, Hansen EB, Nilsson B. Prediction of reversible IgG1 aggregation occurring in a size exclusion chromatography column is enabled through a model based approach. *Biotechnol J.* 2015;10:1814–1821.
6. Otero B, Degerman M, Hansen TB, Hansen EB, Nilsson B. Model-based design and integration of a two-step

- biopharmaceutical production process. *Bioproc Biosyst Eng.* 2014;37:1989–1996.
7. Sellberg A, Ojala F, Nilsson B. Model-based comparison of antibody dimerization in continuous and batch-wise downstream processing. *Antibodies.* 2015;4:157–169.
 8. Godawat R, Brower K, Jain S, Konstantinov K, Riske F, Warikoo V. Periodic counter-current chromatography—design and operational considerations for integrated and continuous purification of proteins. *Biotechnol J.* 2012;7:1496–1508.
 9. Nestola P, Silva RJS, Peixoto C, Alves PM, Carrondo MJT, Mota JPB. Robust design of adenovirus purification by two-column, simulated moving-bed, size-exclusion chromatography. *J Biotechnol.* 2015;213:109–119.
 10. Steinebach F, Karst D, Morbidelli M. *Continuous Production of Proteins: Integration of Polishing Using MCSGP.* In: Conference on Integrated Continuous Biomanufacturing II, Berkeley; 2015.
 11. Tiainen P, Skibstrup D, Ahmedian H, Bjelke JR. *Continuous Protein Chromatography—Continuous Capture Coupled with Two-Step Automated Polish.* In: Conference on Integrated Continuous Biomanufacturing I, Barcelona; 2013.
 12. Nilsson B, Andersson N, Holmqvist A, Sellberg A, Staby A. *Straight Through Processing using Integrated Chromatography Columns Sequences.* In: Conference on Recovery of Biological Products XVII, Bermuda; 2016.
 13. Tiainen P, Skibstrup D, Ahmedian H, Bjelke JR. *Modification of An ÄKTA Enabling Automation of Multi-Step and Complex Protein Purification Processes.* In: Conference on Integrated Continuous Biomanufacturing I, Barcelona; 2013.
 14. Hansen E, Møllerup JM. The influence of retention on the plate height in ion-exchange chromatography. *Sep Sci Technol.* 2005; 39:2011–2030.
 15. Schmidt-Traub H. *Preparative Chromatography of Fine Chemicals and Pharmaceutical Agents.* Weinheim, Germany: Wiley-VCH; 2006.
 16. Biegler LT. *Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes.* Philadelphia, PA: SIAM; 2010.
 17. Gass SI. *Linear Programming: Methods and Applications*, 5 ed. New York, NY: Dover Publications; 2003.
 18. Python Software Foundation. *Python Programming Language – Official Website*; 2009. Available at: www.python.org, accessed on August 30, 2016.
 19. Kraft D. *A Software Package for Sequential Quadratic Programming.* Köln, Germany: Deutsche Forschungs- und Versuchungsanstalt für Luft- und Raumfahrt e.V.; 1988.
 20. Jones E, Oliphant T, Peterson P, et al. *SciPy: Open source scientific tools for Python*; 2001. Available at: <http://www.scipy.org/>, accessed on August 30, 2016.
 21. Oliphant TE. Python for scientific computing. *Comput Sci Eng.* 2007;9:10–20.
 22. Holmqvist A, Sellberg A. *A Generic PAT Software Interface for On-Line Monitoring and Control of Chromatographic Separation Systems.* In: European Symposium on Computer-Aided Process Engineering, Portorož, Slovenia; June 12–15, 2016.
 23. Zheng L, Nakagawa H. *OPC (OLE for process control) Specification and Its Developments.* In: Proceedings of the 41st SICE Annual Conference, Osaka, Japan; August 5–7, 2002.
 24. Andersson N, Lövgren A, Olofsson M, Sellberg A, Nilsson B. *The Orbit Controller.* Lund, Sweden: Lund University; 2016.
 25. Phillips D. *Python 3 Object Oriented Programming.* Birmingham, UK: Packt Publishing Ltd; 2010.
 26. Andersson C, Führer C, Åkesson J. Assimulo: a unified framework for ODE solvers. *Math Comput Simul.* 2015;116:26–43.
 27. Hindmarsh AC, Brown PN, Grant KE, Lee SL, Serban R, Shumaker DE, Woodward CS. SUNDIALS: suite of nonlinear and differential/algebraic equation solvers. *ACM Trans Math Softw.* 2005;31:363–396.
 28. Melander WR, El Rassi Z, Horváth C. Interplay of hydrophobic and electrostatic interactions in biopolymer chromatography. Effect of salts on the retention of proteins. *J Chromatogr A.* 1989;469:3–27.
 29. Ojala F, Max-Hansen M, Kifle D, Borg N, Nilsson B. Modelling and optimisation of preparative chromatographic purification of europium. *J Chromatogr A.* 2012;1220:21–25.

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